

Unit: 2 High Frequency Transmission Lines.

Total line voltage, current.

$$V = ae^{px} + be^{-px}$$

$$I = ce^{px} + de^{-px}$$

$$p = \alpha + j\beta = 0 + j\beta = j\beta$$

In high freq,
freq $\rightarrow \uparrow$
amplitude constant
 $\alpha \rightarrow$ " "

no change in α . so negligible.

HF condition $V = ae^{j\beta x} + be^{-j\beta x}$, $I = ce^{j\beta x} + de^{-j\beta x}$

Total line voltage, current in terms of sending end

$\omega L \gg R$ $V = V_s \cosh(p\alpha) - Z_0 I_s \sinh(p\alpha)$

$\omega C \gg G$ $V = V_s \cosh(j\beta x) - Z_0 I_s \sinh(j\beta x)$

$\alpha = 0$

$\sqrt{RG} = 0$
 $R = 0, G = 0$

$$V = V_s \cos(\beta x) - j Z_0 I_s \sin(\beta x)$$

$$I = I_s \cosh(p\alpha) - \frac{V_s}{Z_0} \sinh(p\alpha)$$

$$I = I_s \cos(\beta x) - j \frac{V_s}{Z_0} \sin(\beta x) \rightarrow (2)$$

Total line voltage, current in terms of receiving end.

$$V = V_R \cosh[p(1-x)] + I_R Z_0 \sinh[p(1-x)]$$

$$V = V_R \cosh[j\beta(1-x)] + I_R Z_0 \sinh[j\beta(1-x)]$$

$$V = V_R \cos[\beta(1-x)] + j I_R Z_0 \sin[\beta(1-x)] \rightarrow (3)$$

$$I = I_R \cosh p(1-x) + \frac{V_R}{Z_0} \sinh p(1-x)$$

$$I = I_R \cos[\beta(1-x)] + j \frac{V_R}{Z_0} \sin[\beta(1-x)] \rightarrow (4)$$

To find sending end voltage and current.

$$V_s = V_R \cosh pl + I_R Z_0 \sinh pl$$

$$V_s = V_R \cosh(j\beta l) + I_R Z_0 \sinh(j\beta l)$$

$$V_s = V_R \cos(\beta l) + j I_R Z_0 \sin(\beta l) \rightarrow (5)$$

$$\hat{I}_S = \hat{I}_R \cosh \beta l + \frac{V_R}{Z_0} \sinh \beta l$$

$$\hat{I}_S = \hat{I}_R \cos(\beta l) + j \frac{V_R}{Z_0} \sin(\beta l) \rightarrow (6)$$

To find receiving end voltage and current.

$$V_R = V_S \cosh \beta l - Z_0 \hat{I}_S \sinh \beta l$$

$$V_R = V_S \cos(\beta l) - j Z_0 \hat{I}_S \sin(\beta l) \rightarrow (7)$$

$$\hat{I}_R = \hat{I}_S \cosh \beta l - \frac{V_S}{Z_0} \sinh \beta l$$

$$\hat{I}_R = \hat{I}_S \cos(\beta l) - j \frac{V_S}{Z_0} \sin(\beta l) \rightarrow (8) \quad \left[\frac{Z_R \cosh \beta l + Z_0 \sinh \beta l}{Z_0 \cosh \beta l + Z_R \sinh \beta l} \right]$$

Input impedance:

$$Z_S = \left[\frac{V_R \cosh \beta l + \hat{I}_R Z_0 \sinh \beta l}{\hat{I}_R Z_0 \cosh \beta l + V_R \sinh \beta l} \right] Z_0$$

$$Z_S = Z_0 \left[\frac{Z_R \cos(\beta l) + j Z_0 \sin(\beta l)}{Z_0 \cos(\beta l) + j Z_R \sin(\beta l)} \right]$$

$$Z_S = Z_0 \left[\frac{Z_R + j Z_0 \tan \beta l}{Z_0 + j Z_R \tan \beta l} \right]$$

$$= \left[\frac{V_R \cosh(j\beta l) + \hat{I}_R Z_0 \sinh(j\beta l)}{\hat{I}_R Z_0 \cosh(j\beta l) + V_R \sinh(j\beta l)} \right] Z_0 \quad \text{take } \cos(\beta l) \text{ common.}$$

$$\hat{I}_S = \left[\frac{V_R \cos(\beta l) + j \hat{I}_R Z_0 \sin(\beta l)}{\hat{I}_R Z_0 \cos(\beta l) + j V_R \sin(\beta l)} \right] Z_0 \rightarrow (9)$$

$$\begin{aligned} Z_S &= Z_0 \left[\frac{(Z_R + Z_0)e^{\beta l} + (Z_R - Z_0)e^{-\beta l}}{(Z_R + Z_0)e^{\beta l} - (Z_0 - Z_R)e^{-\beta l}} \right] \\ &= Z_0 \left[\frac{(Z_R + Z_0)e^{j\beta l} + (Z_R - Z_0)e^{-j\beta l}}{(Z_R + Z_0)e^{j\beta l} - (Z_0 - Z_R)e^{-j\beta l}} \right] \end{aligned}$$

$$Z_s = Z_0 \left[\frac{1 + k e^{-j^2 \beta l}}{1 - k e^{-j^2 \beta l}} \right] \rightarrow (10)$$

Transfer impedance:

$$I_t = \frac{V_s}{I_R} = Z_R \cosh \beta l + Z_0 \sinh \beta l$$

$$Z_t = Z_R \cosh(j\beta l) + Z_0 \sinh(j\beta l)$$

$$Z_t = Z_R \cos(\beta l) + j Z_0 \sin(\beta l) \rightarrow (11)$$

open circuit and short circuit lines:

$$Z_{oc} = Z_0 \coth \beta l = Z_0 \left[\frac{\cosh \beta l}{\sinh \beta l} \right]$$

$$= Z_0 \left[\frac{\cosh(j\beta l)}{\sinh(j\beta l)} \right]$$

$$= Z_0 \left[\frac{\cos(\beta l)}{j \sin(\beta l)} \right]$$

$$Z_{oc} = Z_0 \cot \beta l \left(\frac{1}{j} \right)$$

$$Z_{oc} = -j Z_0 \cot \beta l \rightarrow (12)$$

$$Z_{sc} = Z_0 \left[\frac{\sinh \beta l}{\cosh \beta l} \right]$$

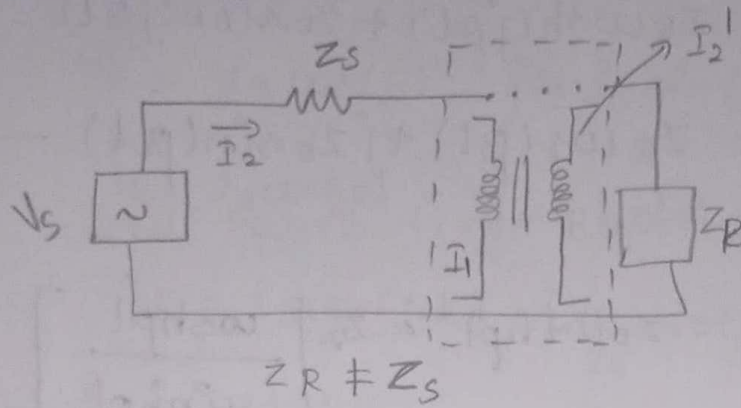
$$= Z_0 \left[\frac{\sinh(j\beta l)}{\cosh(j\beta l)} \right]$$

$$= Z_0 j \frac{\sin \beta l}{\cos \beta l}$$

$$Z_{sc} = j Z_0 \tan \beta l \rightarrow (13)$$

Reflection Factor:

It is defined as the ratio of current actually flowing in the load to the current which might flow under maximum matched condition.



$$Z_R = Z_s$$

$$\frac{I_2'}{I_1} = \sqrt{\frac{Z_s}{Z_R}} \quad Z_R = Z_s$$

$$I_1 = \frac{V_s}{Z_s + Z_R}$$

$$I_1 = \frac{V_s}{2Z_s}$$

$$I_2' = I_1 \sqrt{\frac{Z_s}{Z_R}}$$

$$I_2' = \frac{V_s}{2Z_s} \sqrt{\frac{Z_s}{Z_R}}$$

$$I_2' = \frac{V_s}{2\sqrt{Z_s Z_R}}$$

with mapping
matched

$$I_2 = \frac{|V_s|}{|Z_s + Z_R|}$$

→ without mapping
unmatched.

$$K = \frac{|\bar{I}_2|}{|\bar{I}_1|}$$

$$= \frac{V_s}{\frac{2\sqrt{Z_s Z_R}}{V_s}} = \frac{|2\sqrt{Z_s Z_R}|}{|Z_s + Z_R|}$$

To find
w/L - why we use
ln()

Reflection loss - reciprocal of reflection factor.

$$\text{Reflection loss} = \ln \left(\frac{1}{\text{Reflection factor}} \right)$$

1. The transmission line 10km long is ^{perfectly} terminated _{matched} properly at the far end. At a freq. of 1kHz, a attenuation and phase constant of the line are respectively 0.03 Np/km and 0.03 rad/km. If the far end voltage at 1kHz is 4∠0° V. calculate sending end voltage of the line.

soln:

$$l = 10\text{km}, f = 1\text{kHz}, p = \alpha + j\beta = 0.03 + j0.03$$

$$V_R = 4\angle 0^\circ \text{ V}$$

properly terminated

$$Z_s = Z_R = Z_0$$

$$V_s = V_R \cosh pl + \bar{I}_R Z_0 \sinh pl$$

$$V_s = V_R \cosh pl + V_R \sinh pl$$

$$V_s = V_R [\cosh pl + \sinh pl]$$

$$\begin{cases} V_R = \bar{I}_R Z_R \\ = \bar{I}_R Z_0 \end{cases}$$

$$\cosh pl = \cosh(0.3 + j0.3)$$

$$= \cosh(0.3) \cosh(j0.3) + \sinh(0.3) \sinh(j0.3)$$

$$= \cosh(0.3) \cos(0.3 \times 57.3^\circ) + j \sinh(0.3) \sin(0.3 \times 57.3^\circ)$$

$$= 1.0453 (0.9553) + j(0.3045)(0.2955) \quad 57.3^\circ$$

$$\cosh \beta l = 0.9985 + j0.08997$$

$$\sinh \beta l = \sinh(0.3 + j0.3)$$

$$= \sinh(0.3) \cosh(j0.3) + \cosh(0.3) \sinh(j0.3)$$

$$= \sinh(0.3) \cos(0.3 \times 57.3^\circ) + j \cosh(0.3) \sin(0.3 \times 57.3^\circ)$$

$$= 0.3045(0.9553) + j(1.04533)(0.2955)$$

$$= 0.2908 + j0.3088$$

$$V_s = 4 \angle 0^\circ [0.9985 + j0.08997 + 0.2908 + j0.3088]$$

$$= 4 \angle 0^\circ [1.2893 + j0.3987]$$

$$V_s = 5.1572 + j1.5948 \quad \checkmark$$

$$V_s = V_R [\cosh \beta l + \sinh \beta l] = V_R [e^{\beta l}]$$

$$= V_R [e^{(0.03 + j0.03) \times 10}]$$

$$= V_R [e^{0.3 + j0.3}]$$

$$= V_R [e^{0.3} \cdot e^{j0.3}]$$

$$= (4 \angle 0^\circ) (1.3498) (\cos(0.3 \times 57.3^\circ) + j \sin(0.3 \times 57.3^\circ))$$

$$= 5.3992 (0.9553 + j0.2955)$$

$$V_s = 5.1578 + j1.5954 \quad \checkmark$$

2. Find the sending end impedance of the line having negligible losses, where Z_0 is

55 Ω , receiving end impedance $155 + j75 \Omega$
and line is 1.183 times the wavelength.

soln:

distortionless line, $\rho = j\beta$.

$$\lambda = 1.183 \lambda$$

$$\lambda = 1.183 \left(\frac{2\pi}{\beta} \right)$$

$$\beta = \frac{1.183(2\pi)}{\lambda}$$

$$Z_s = Z_0 \left[\frac{Z_R + jZ_0 \tan \beta l}{Z_0 + jZ_R \tan \beta l} \right]$$

$$= Z_0 \left[\frac{Z_R + jZ_0 \tan \left[\frac{1.183(2\pi)}{\lambda} \lambda \right]}{Z_0 + jZ_R \tan \left[\frac{1.183(2\pi)}{\lambda} \lambda \right]} \right]$$

$$= 55 \left[\frac{155 + j75 + j55 \tan(1.183 \times 2\pi \times 57.3)}{55 + j(155 + j75) \tan(1.183 \times 2\pi \times 57.3)} \right]$$

$$= 55 \left[\frac{155 + j75 + j121.7857}{55 + (-75 + j155) \tan(1.183 \times 2\pi \times 57.3)} \right]$$

$$= 55 \left[\frac{155 + j196.7857}{55 - 166.0714 + j343.2143} \right]$$

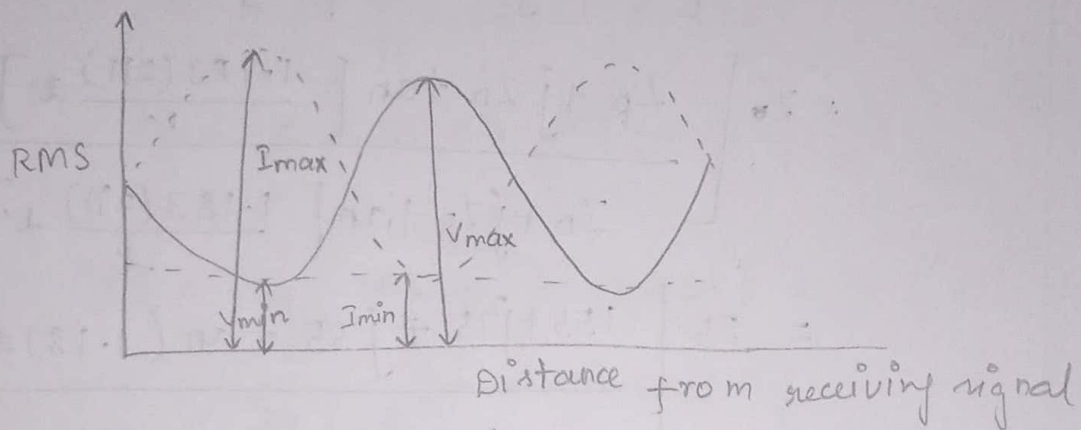
$$= 55 \left[\frac{155 + j196.7857}{-111.0714 + j343.2143} \right]$$

$$= 55 (0.3867 - j0.5767)$$

$$Z_s = 21.2685 - j31.7185 \Omega$$

Standing wave ratio: [SWR]

The reflection takes place when the line is not terminated in its characteristic impedance. It causes reflection of waves (i.e. mismatch). The combination of incident and reflected waves gives rise to standing wave pattern, the standing waves of current and voltage with definite maxima and minima along with the line.



The incident wave and reflected wave are in phase where the resultant of signal is maximum.

$$V_{\max} = V_i + V_r$$

$$I_{\max} = I_i + I_r$$

[$V_i \rightarrow$ incident voltage]

$V_r \rightarrow$ reflected voltage]

The incident and reflected waves are out-of-phase where the resultant of signal is minimum.

$$V_{\min} = V_i - V_r$$

$$I_{\min} = I_i - I_r$$

The ratio of maximum and minimum magnitude of current or voltage on a line having a standing wave is called standing wave Ratio [SWR].

$$S = \frac{|V_{\max}|}{|V_{\min}|} = \frac{|I_{\max}|}{|I_{\min}|}$$

For perfectly matched, $S=1$

For unperfectly matched, $S > 1$ [$\because V_{\min}$ or $I_{\min}$ is much less].

$$S = \frac{|V_i + V_r|}{|V_i - V_r|}$$

$$= \frac{V_i \left[1 + \left| \frac{V_r}{V_i} \right| \right]}{V_i \left[1 - \left| \frac{V_r}{V_i} \right| \right]}$$

$$S = \frac{1 + |k|}{1 - |k|}$$

$$VSWR = \frac{1 + |k|}{1 - |k|}$$

$$CSWR = \frac{1 - |k|}{1 + |k|}$$

$$(1 - |k|) S = 1 + |k|$$

$$S - |k| S = 1 + |k|$$

$$S - 1 = |k| + |k| S$$

$$= |k| [1 + S]$$

$$|k| = \frac{S - 1}{S + 1}$$

$$\text{Put } S = \frac{V_{\max}}{V_{\min}}$$

$$|k| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$$

Nodes - $V_{\max}, V_{\min}$

Power and Impedance measurement of Line:

At maximum voltage and current, the incident and reflected waves are in phase.

$$V_{\max} = \frac{I_R (Z_R + Z_0)}{2} (1 + |k|)$$

$$I_{\max} = \frac{I_R (Z_R + Z_0)}{2 Z_0} (1 + |k|)$$

$$V_{\min} = \frac{I_R (Z_R + Z_0)}{2} (1 - |k|)$$

$$I_{\min} = \frac{I_R (Z_R + Z_0)}{2 Z_0} (1 - |k|)$$

At minimum voltage and current, incident and reflected waves are out-of-phase.

The ratio of $V_{\max}$ and $I_{\max}$

$$\frac{V_{\max}}{I_{\max}} = \frac{\frac{I_R (Z_R + Z_0)}{2} (1 + |k|)}{\frac{I_R (Z_R + Z_0)}{2 Z_0} (1 + |k|)} = Z_0$$

The ratio of $V_{\min}$ and $I_{\min}$

$$\frac{V_{\min}}{I_{\min}} = Z_0$$

The characteristic impedance

$$Z_0 = \frac{V_{\max}}{I_{\max}} = \frac{V_{\min}}{I_{\min}}$$

$$I_{\max} = \frac{V_{\max}}{I_{\min}} = \frac{\frac{I_R (Z_R + Z_0)}{2} (1 + |k|)}{\frac{I_R (Z_R + Z_0)}{2 Z_0} (1 - |k|)} = \frac{Z_0 (1 + |k|)}{1 - |k|}$$

$$Z_{\max} = Z_0 S$$

$$Z_{\min} = \frac{V_{\min}}{I_{\max}} = \frac{\frac{I_R (Z_R + Z_0)}{2} (1 - |k|)}{\frac{I_R (Z_R + Z_0)}{2 Z_0} (1 + |k|)}$$

$$Z_{\min} = \frac{Z_0}{S}$$

$$P_{\max} = \frac{V_{\max}^2}{Z_{\max}} = I_{\max}^2 Z_{\max} \rightarrow (1)$$

$$P_{\min} = \frac{V_{\min}^2}{Z_{\min}} = I_{\min}^2 Z_{\min} \rightarrow (2)$$

$$(1) \times (2) \Rightarrow P \times P = I_{\max}^2 Z_{\max} \cdot I_{\min}^2 Z_{\min}$$

$$P^2 = I_{\max}^2 I_{\min}^2 Z_{\max} Z_{\min} \rightarrow (3)$$

(or)

$$\text{Total power} \left\{ \begin{aligned} P_T^2 &= \frac{V_{\max}^2 V_{\min}^2}{Z_{\max} Z_{\min}} \rightarrow (4) \end{aligned} \right.$$

$$P_T^2 = \frac{V_{\max}^2 V_{\min}^2}{Z_0 S \cdot \frac{Z_0}{S}} = \frac{V_{\max}^2 \cdot V_{\min}^2}{Z_0^2}$$

$$P_T = \frac{V_{\max} V_{\min}}{Z_0} \rightarrow (5)$$

$$(5) \Rightarrow P^2 = I_{\max}^2 I_{\min}^2 Z_0 S \cdot \frac{Z_0}{S} = I_{\max}^2 I_{\min}^2 Z_0^2$$

$$P = I_{\max} I_{\min} Z_0 \rightarrow (6)$$

1. In a transmission line ^{VSWR} voltage SWR is given by 2.5. The characteristic impedance is

50Ω . The line is to transmit the power of $25W$. Determine maximum and min. voltage and current.

soln:

$$Z_0 = 50\Omega, P = 25W$$

$$V_{max}, V_{min}, I_{max}, I_{min} = ?$$

$$S = \frac{|V_{max}|}{|V_{min}|} = 2.5$$

$$V_{max} = 2.5 V_{min}$$

$$P_T = \frac{V_{max} V_{min}}{Z_0}$$

$$25 = \frac{2.5 V_{min}^2}{50}$$

$$V_{min}^2 = 500$$

$$V_{min} = 22.3607 V$$

$$V_{max} = 55.9017 V$$

$$\frac{V_{min}}{I_{min}} = Z_0$$

$$I_{min}$$

$$I_{min} = \frac{V_{min}}{Z_0} = 0.4472 A$$

$$\frac{V_{max}}{I_{max}} = Z_0$$

$$I_{max} = \frac{V_{max}}{Z_0} = 1.1180 A$$

Don't use the formula

$$P = I_{max}^2 Z_{max}$$

$$P = I_{min}^2 Z_{max}$$

Verify:

$$P = I_{max} I_{min} Z_0$$

$$= (1.1180)(0.4472)(50)$$

$$= 24.998$$

$$P \approx 25W \checkmark$$

2. A line with zero dissipation has $L = 2.5 \mu H/m$ and $C = 4.45 pF/m$. If the line is operated at $10 MHz$, find characteristic impedance, phase

constant, velocity of propagation, wavelength.

soln:

$$Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{2.5 \times 10^{-6}}{4.45 \times 10^{-12}}} = 0.7495 \times 10^3$$

$$Z_0 = 749.5 \Omega \checkmark$$

$$\beta = \omega \sqrt{LC}$$

$$= 2\pi f \sqrt{LC}$$

$$= 2\pi \times 10 \times 10^6 \sqrt{2.5 \times 10^{-6} \times 4.45 \times 10^{-12}}$$

$$= 209.4641 \times 10^6 \times 10^{-9}$$

$$= 209.4641 \times 10^{-3}$$

$$\beta = 0.2094 \text{ rad/metre} \checkmark$$

$$v = \frac{\omega}{\beta} = \frac{2\pi \times 10 \times 10^6}{0.2094} = 299.9 \times 10^6$$

$$v = 2.999 \times 10^8 \text{ m/s} \checkmark$$

$$\lambda = \frac{2\pi}{\beta} = 29.99 \text{ m} \checkmark$$

3. A transmission line of characteristic impedance 600Ω is terminated by a reactance of $j150 \Omega$, find input impedance of a section 25 cm long at a freq. of 300 MHz .

soln:

$$\lambda = 25 \text{ cm}, f = 300 \text{ MHz}, Z_0 = 600 \Omega,$$

$$Z_R = 0 + j150 \Omega.$$

$$k = \frac{Z_R - Z_0}{Z_R + Z_0}$$

$$k = \frac{j150 + 600}{j150 + 600} = -0.8823 + j0.4705$$

$$\beta l = \frac{2\pi}{\lambda} l ; \lambda = \frac{c}{f} = \frac{3 \times 10^8}{300 \times 10^6} = 1 \text{ m}$$

$$\beta l = \frac{2\pi}{1} (0.25) = \frac{\pi}{2}$$

$$Z_{in} = Z_0 \left[\frac{1 + k e^{-j2\beta l}}{1 - k e^{-j2\beta l}} \right]$$

$$e^{-j2\beta l} = e^{-j2\frac{\pi}{2}} = e^{-j\pi} = \cos \pi - j \sin \pi = -1$$

$$\begin{aligned} Z_{in} &= 600 \left[\frac{1 + (-0.8823 + j0.4705)(-1)}{1 - (-0.8823 + j0.4705)(-1)} \right] \\ &= (7.5 \times 10^{-4} - j4) 600 = 0.45 - j2400.26 \end{aligned}$$

4. A transmission line with $Z_0 = 300 \Omega$ is terminated in a purely resistive load. It is found by measurement that the minimum value of voltage upon it is $5 \mu V$ and the maximum value is $7.5 \mu V$. Calculate the value of load resistance.

soln:

$$Z_R = ?$$

$$Z_0 = 300 \Omega, V_{min} = 5 \mu V, V_{max} = 7.5 \mu V.$$

$$S = \left| \frac{V_{max}}{V_{min}} \right| = \frac{7.5 \mu V}{5 \mu V} = 1.5$$

$$k = \frac{S-1}{S+1} = 0.20$$

$$k = \frac{Z_R - Z_0}{Z_R + Z_0}$$

$$0.2 = \frac{Z_R - 300}{Z_R + 300}$$

$$0.2 Z_R + 60 = Z_R - 300$$

$$360 = 0.8 Z_R$$

$$Z_R = 450 \Omega$$

$$Z_S = Z_0 \left[\frac{Z_R + j Z_0 \tan \beta l}{Z_0 + j Z_R \tan \beta l} \right]$$

$$= Z_0 \left[\frac{Z_R + j Z_0 \tan \frac{\pi}{2}}{Z_0 + j Z_R \tan \frac{\pi}{2}} \right] = \frac{Z_0 \left[\tan \frac{\pi}{2} \right]}{\left[\tan \frac{\pi}{2} \right]} \left[\frac{0 + j Z_0}{0 + j Z_R} \right]$$

$$= \frac{Z_0^2}{Z_R} = \frac{(600)^2}{j 150} = -j 2400.$$